Compte rendu – TP Cisco VLAN et Routage Inter-VLAN

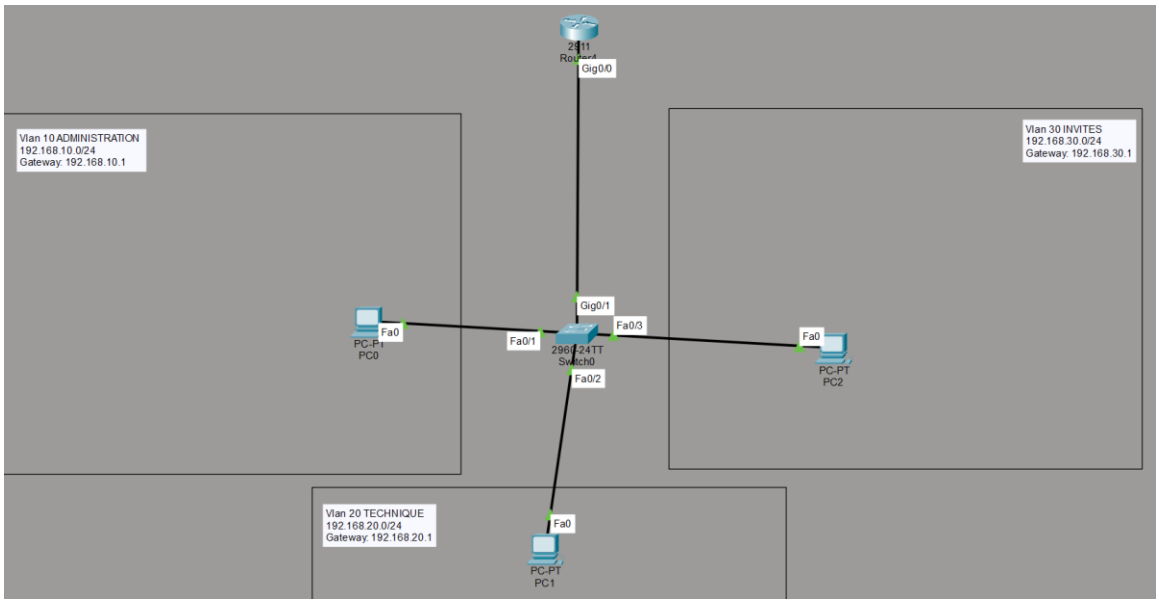
1. Contexte

Dans le cadre de ce TP, j'ai été chargé de mettre en place une segmentation réseau pour une PME afin d'améliorer la sécurité et la gestion des flux.

2. Objectifs

- Mettre en place des VLAN
- Configurer le routage inter-VLAN
- Sécuriser le réseau avec une ACL

3. Topologie réseau



4. Plan d'adressage

VLAN 10 – ADMIN – 192.168.10.0/24
VLAN 20 – TECH – 192.168.20.0/24
VLAN 30 – GUEST – 192.168.30.0/24

5. Configuration du switch

Création des VLAN et affectation des ports.

```
Switch#sho
Switch#show vlan
Switch#show vlan bri
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/2
10   ADMINISTRATION                   active    Fa0/1
20   TECHNIQUE                        active    Fa0/2
30   INVITES                          active    Fa0/3
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

TRUNK

```
Switch# show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      1-1005

Port        Vlans allowed and active in management domain
Gig0/1      1,10,20,30

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      1,10,20,30

Switch#
```

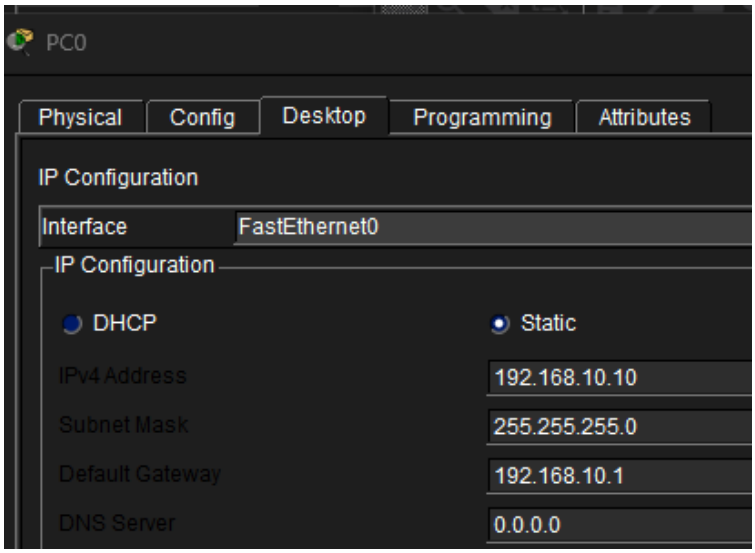
6. Configuration du routeur

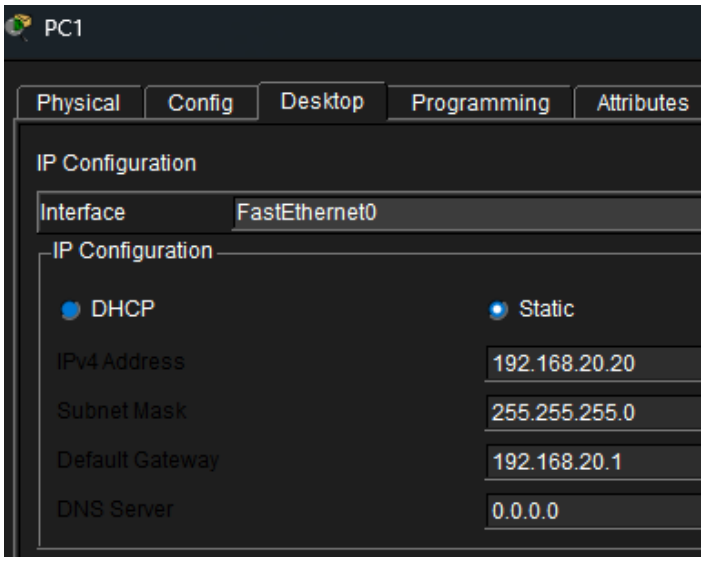
Configuration des sous-interfaces et routage inter-VLAN.

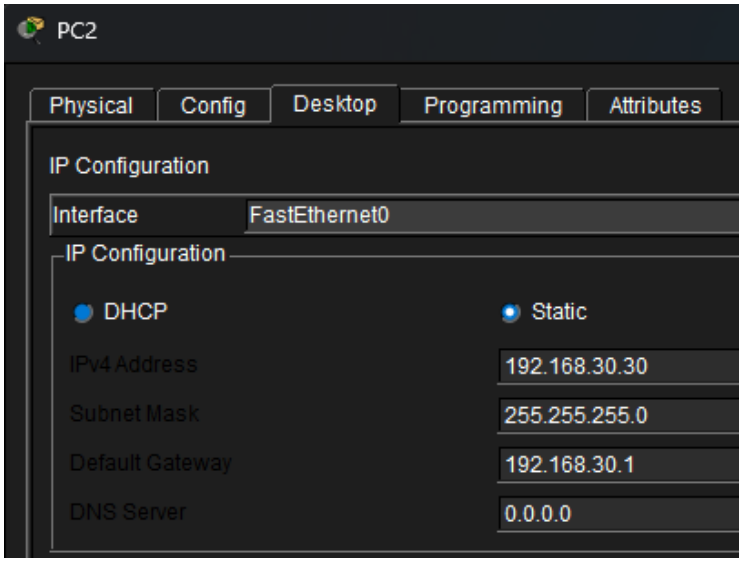
```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  192.168.10.1    YES manual up                    up
GigabitEthernet0/0.20  192.168.20.1    YES manual up                    up
GigabitEthernet0/0.30  192.168.30.1    YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
!
interface GigabitEthernet0/0
 no ip address
 duplex auto
 speed auto
!
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
 ip access-group 100 in
!
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
!
interface GigabitEthernet0/0.30
 encapsulation dot1Q 30
 ip address 192.168.30.1 255.255.255.0
 ip access-group 100 in
!
interface GigabitEthernet0/1
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
ip classless
!
ip flow-export version 9
!
```

7. Configuration des postes

Configuration des adresses IP.







8. Tests de connectivité

[CAPTURE : ping ADMIN → TECH]

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::201:42FF:FEA1:AB11
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.10.10
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.10.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 192.168.20.20

Pinging 192.168.20.20 with 32 bytes of data:

Reply from 192.168.20.20: bytes=32 time<1ms TTL=127
Reply from 192.168.20.20: bytes=32 time<1ms TTL=127
Reply from 192.168.20.20: bytes=32 time<1ms TTL=127
Reply from 192.168.20.20: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

[CAPTURE : ping TECH → GUEST]

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::20C:85FF:FEE0:38B5
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.20.20
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.20.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 192.168.30.30

Pinging 192.168.30.30 with 32 bytes of data:

Reply from 192.168.30.30: bytes=32 time<1ms TTL=127
Reply from 192.168.30.30: bytes=32 time<1ms TTL=127
Reply from 192.168.30.30: bytes=32 time<1ms TTL=127
Reply from 192.168.30.30: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

[CAPTURE : ping GUEST → ADMIN]

```
C:\>
IPCONFIG

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::2E0:F9FF:FEA3:66D7
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.30.30
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.30.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

9. Sécurisation (ACL)

Mise en place d'une ACL pour restreindre l'accès.

```
Router#show acc
Router#show access-lists
Extended IP access list 100
    10 deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255 (8 match(es))
    20 permit ip any any (33 match(es))
```

```
interface GigabitEthernet0/1
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
ip classless
!
ip flow-export version 9
!
!
access-list 100 deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
access-list 100 permit ip any any
!
!
!
```

10. Résultats

Les VLANs sont fonctionnels, le routage inter-VLAN est opérationnel et les restrictions sont respectées.

11. Difficultés rencontrées

Lors de la mise en place d'une liste de contrôle d'accès (ACL 100) visant à interdire le trafic ICMP (Ping) du VLAN 30 (Source) vers le VLAN 10 (Destination), le filtrage ne fonctionnait pas. Le trafic restait autorisé malgré la configuration correcte de la règle de rejet (deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255).

L'ACL était appliquée sur l'interface de destination (GigabitEthernet 0/0.10) avec la direction IN.

1. Erreur de sens : Dans la logique de routage, un paquet provenant du VLAN 30 entre par l'interface 30 et sort par l'interface 10. Appliquer une règle en IN sur l'interface 10 ne permettait pas d'intercepter le flux sortant.

2. Erreur de placement : Une ACL étendue doit, par convention d'optimisation, être placée au plus près de la source pour éviter que le routeur n'utilise des ressources (CPU/Table de routage) pour un paquet destiné à être rejeté.

Pour corriger l'incident, l'ACL a été déplacée sur l'interface du VLAN 30 en entrée :

interface gigabitEthernet 0/0.30

ip access-group 100 in

Cet incident a permis de consolider la règle d'or du filtrage réseau :

- ACL Étendue : Toujours au plus proche de la source en direction IN.
- ACL Standard : Toujours au plus proche de la destination en direction OUT.

12. Conclusion

Ce TP m’a permis de comprendre l’importance de la segmentation réseau et des mécanismes de sécurité associés.